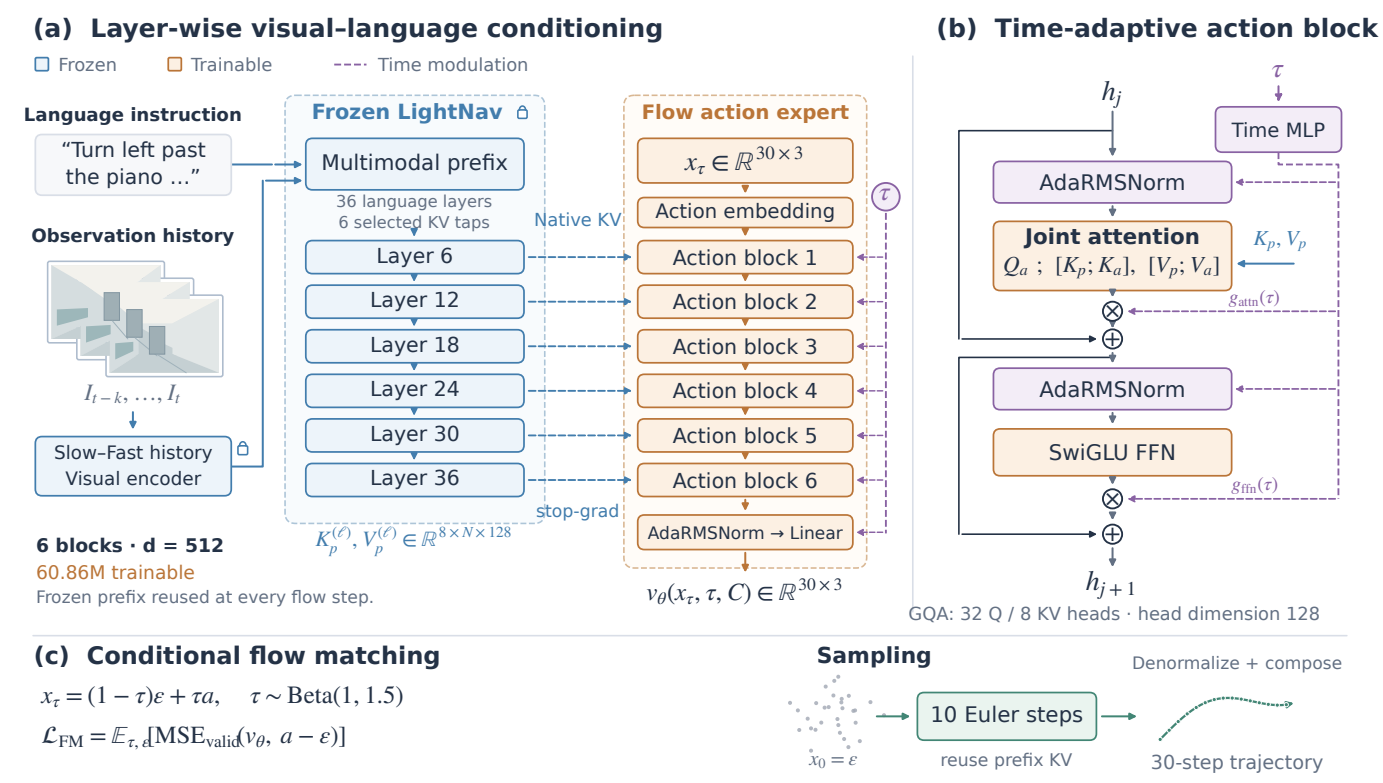

(a) Layer-wise visual–language conditioning
Frozen
Trainable
Time modulation
Language instruction
"Turn left past the piano …"
Observation history
$I_{t-k}, \ldots, I_t$
Slow–Fast history Visual encoder
Frozen LightNav
Multimodal prefix
36 language layers
6 selected KV taps
Layer 6
Layer 12
Layer 18
Layer 24
Layer 30
Layer 36
Native KV
stop-grad
$K_p^{(\ell)}, V_p^{(\ell)} \in \mathbb{R}^{8\times N\times 128}$
Flow action expert
$x_\tau \in \mathbb{R}^{30\times 3}$
Action embedding
Action block 1
Action block 2
Action block 3
Action block 4
Action block 5
Action block 6
AdaRMSNorm → Linear
τ
$v_\theta(x_\tau, \tau, C) \in \mathbb{R}^{30\times 3}$
6 blocks · d = 512
60.86M trainable
Frozen prefix reused at every flow step.
(b) Time-adaptive action block
h_j
τ
Time MLP
AdaRMSNorm
Joint attention
Q_a ; $[K_p; K_a]$, $[V_p; V_a]$
K_p, V_p
$g_{\text{attn}}(\tau)$
AdaRMSNorm
SwiGLU FFN
$g_{\text{ffn}}(\tau)$
h_{j+1}
GQA: 32 Q / 8 KV heads · head dimension 128
(c) Conditional flow matching
$x_\tau = (1-\tau)\varepsilon + \tau a, \quad \tau \sim \text{Beta}(1, 1.5)$
$\mathcal{L}_{\text{FM}} = \mathbb{E}_{\tau,\varepsilon}[\text{MSE}_{\text{valid}}(v_\theta, a-\varepsilon)]$
Sampling
$x_0 = \varepsilon$
10 Euler steps
reuse prefix KV
Denormalize + compose
30-step trajectory